

21. Let $A = [a_{ij}] = \begin{bmatrix} \log_5 128 & \log_4 5 \\ \log_5 8 & \log_4 25 \end{bmatrix}$. If A_{ij} is the cofactor of a_{ij} , $C_{ij} = \sum_{k=1}^2 a_{ik} A_{jk}$, $1 \leq i, j \leq 2$ and $C = [C_{ij}]$ then $8|C|$ is equal to :

माना कि $A = [a_{ij}] = \begin{bmatrix} \log_5 128 & \log_4 5 \\ \log_5 8 & \log_4 25 \end{bmatrix}$ है। यदि A_{ij} , a_{ij} का सहखण्ड है, $C_{ij} = \sum_{k=1}^2 a_{ik} A_{jk}$, $1 \leq i, j \leq 2$ और $C = [C_{ij}]$

है, तो $8|C|$ बराबर है :

- (A) 288 (B) 222 (C) 262 (D) 242

Ans. D

Sol. $A = \begin{bmatrix} \log_5 128 & \log_4 5 \\ \log_5 8 & \log_4 25 \end{bmatrix}$

$$C_{ij} = \sum_{k=1}^2 a_{ik} A_{jk}, 1 \leq i, j \leq 2$$

$$C_{11} = \sum_{k=1}^2 a_{1k} A_{1k} = a_{11} A_{11} + a_{12} A_{12}$$

$$C_{11} = \log_5 128 \times \log_4 25 - \log_4 5 \log_5 8$$

$$= 7 \log_5 2 \times \log_2 5 - \frac{1}{2} \log_2 5 \times 3 \log_5 2$$

$$= 7 - \frac{3}{2} = \frac{11}{2}$$

$$C_{12} = \sum_{k=1}^2 a_{1k} A_{2k} = a_{11} A_{21} + a_{12} A_{22}$$

$$= -\log_5 128 \log_4 5 + \log_4 5 \times \log_5 128$$

$$= 0$$

$$C_{21} = \sum_{k=1}^2 a_{2k} A_{1k} = a_{21} A_{11} + a_{22} A_{12}$$

$$= \log_5 8 \log_4 25 - \log_4 25 \log_5 8$$

$$= 0$$

$$C_{22} = \sum_{k=1}^2 a_{2k} A_{2k} = a_{21} A_{21} + a_{22} A_{22}$$

$$= -\log_5 8 \log_4 5 + \log_4 25 \times \log_5 128$$

$$= -\frac{3}{2} \log_5 2 \log_2 5 + \log_2 5 \times 7 \log_5 2$$

$$= 7 - \frac{3}{2} = \frac{11}{2}$$

$$C = \begin{bmatrix} \frac{11}{2} & 0 \\ 0 & \frac{11}{2} \end{bmatrix}$$

$$8|C| = 8 \times \frac{121}{4} = 242$$

22. Let $a_1, a_2, a_3, \dots$ be a G.P. of increasing positive numbers. If $a_3 a_5 = 729$ and $a_2 + a_4 = \frac{111}{4}$, then $24(a_1 + a_2 + a_3)$ is equal to :

माना $a_1, a_2, a_3, \dots$ वर्धमान धनात्मक संख्याओं की एक G.P. है। यदि $a_3 a_5 = 729$ और $a_2 + a_4 = \frac{111}{4}$ हैं, तो

$24(a_1 + a_2 + a_3)$ बराबर है :

- (A) 131 (B) 128 (C) 129 (D) 130

Ans. C

Sol. $ar^2 \cdot ar^4 = 729$ $ar + ar^3 = \frac{111}{4}$

$ar^3 = 27$ $ar = \frac{3}{4}$

$$\frac{ar^3}{ar} = \frac{27}{3/4} \Rightarrow r = 6$$

$$a = \frac{1}{8} \quad \text{Now } 24(a + ar + ar^2) = 129$$

23. Consider an A.P. of positive integers, whose sum of the first three terms is 54 and the sum of the first twenty terms lies between 1600 and 1800. Then its 11th term is :

धनात्मक पूर्णाकों की एक A.P. पर विचार करें, जिसके पहले तीन पदों का योग 54 है और पहले बीस पदों का योग 1600 और 1800 के बीच में है। तब इसका 11 वाँ पद है :

(A) 90

(B) 122

(C) 84

(D) 108

Ans. A

Sol. $a - d + a + a + d = 54$

$$3a = 54 \Rightarrow a = 18$$

$$18 - d, 18, 18 + d$$

$$1600 < \frac{20}{2}(2(18 - d) + 19d) < 1800$$

$$160 < 2(18 - d) + 19d < 180$$

$$160 < 36 + 17d < 180$$

$$124 < 17d < 144$$

$$\frac{124}{17} < d < \frac{144}{17}$$

$$7.29 < d < 8.47$$

$$d = 8$$

$$T_{11} = a + 10d$$

$$= 10 + 10 \times 8 = 90$$

24. Let $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ and $B = I + \text{adj}(A) + (\text{adj } A)^2 + \dots + (\text{adj } A)^{10}$. Then, the sum of all the elements of the matrix B is :

माना $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ तथा $B = I + \text{adj}(A) + (\text{adj } A)^2 + \dots + (\text{adj } A)^{10}$ हो, तो आव्यूह B के सभी अवयवों का योग है :

(A) -110

(B) 22

(C) -124

(D) -88

Ans. D

Sol. $\text{Adj}(A) = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}$

$$(\text{Adj } A)^2 = \begin{bmatrix} 1 & -4 \\ 0 & 1 \end{bmatrix}$$

$$(\text{Adj } A)^{10} = \begin{bmatrix} 1 & -20 \\ 0 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & -4 \\ 0 & 1 \end{bmatrix} + \dots + \begin{bmatrix} 1 & -20 \\ 0 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} 11 & -110 \\ 0 & 11 \end{bmatrix}$$

$\Rightarrow$ sum of elements of B

$$= -88$$

25. If 2 and 6 are the roots of the equation $ax^2 + bx + 1 = 0$, then the quadratic equation, whose roots are $\frac{1}{2a+b}$ and $\frac{1}{6a+b}$, is :

यदि समीकरण $ax^2 + bx + 1 = 0$ के मूल 2 तथा 6 हैं, तो वह द्विघात समीकरण, जिसके मूल $\frac{1}{2a+b}$ तथा $\frac{1}{6a+b}$ हैं, है :

(A) $2x^2 + 11x + 12 = 0$ (B) $x^2 + 8x + 12 = 0$ (C) $4x^2 + 14x + 12 = 0$ (D) $x^2 + 10x + 16 = 0$

Ans. B

Sol. Sum $= 8 = -\frac{b}{a}$

$$\text{Product} = 12 = \frac{1}{a} \Rightarrow a = \frac{1}{12}$$

$$b = -\frac{2}{3}$$

$$2a + b = \frac{2}{12} - \frac{2}{3} = -\frac{1}{2}$$

$$6a + b = \frac{6}{12} - \frac{2}{3} = -\frac{1}{6}$$

$$\text{sum} = -8$$

$$x^2 + 8x + 12 = 0$$

26. Let $A = [a_{ij}]$ be a matrix of order 3×3 , with $a_{ij} = (\sqrt{2})^{i+j}$. If the sum of all the elements in the third row of A^2 is $\alpha + \beta\sqrt{2}$, $\alpha, \beta \in \mathbb{Z}$, then $\alpha + \beta$ is equal to :

माना कि $A = [a_{ij}]$, $a_{ij} = (\sqrt{2})^{i+j}$, 3×3 का एक आव्यूह है। यदि A^2 की तीसरी पंक्ति के सभी अवयवों का योग $\alpha + \beta\sqrt{2}$, $\alpha, \beta \in \mathbb{Z}$

है, तो $\alpha + \beta$ बराबर है :

(A) 168

(B) 210

(C) 224

(D) 280

Ans. C

Sol.
$$A = \begin{bmatrix} (\sqrt{2})^2 & (\sqrt{2})^3 & (\sqrt{2})^4 \\ (\sqrt{2})^3 & (\sqrt{2})^4 & (\sqrt{2})^5 \\ (\sqrt{2})^4 & (\sqrt{2})^5 & (\sqrt{2})^6 \end{bmatrix}$$

$$A = \begin{bmatrix} 2 & 2\sqrt{2} & 4 \\ 2\sqrt{2} & 4 & 4\sqrt{2} \\ 4 & 4\sqrt{2} & 8 \end{bmatrix}$$

$$A^2 = 2^2 \begin{bmatrix} 1 & \sqrt{2} & 2 \\ \sqrt{2} & 2 & 2\sqrt{2} \\ 2 & 2\sqrt{2} & 4 \end{bmatrix} \begin{bmatrix} 1 & \sqrt{2} & 2 \\ \sqrt{2} & 2 & 2\sqrt{2} \\ 2 & 2\sqrt{2} & 4 \end{bmatrix}$$

$$= 4 \begin{bmatrix} - & - & - \\ - & - & - \\ (2+4+8) & (2\sqrt{2}+4\sqrt{2}+8\sqrt{2}) & (4+8+16) \end{bmatrix}$$

Sum of elements of 3rd row = $4(14 + 14\sqrt{2} + 28)$

$$= 4(42 + 14\sqrt{2})$$

$$= 168 + 56\sqrt{2}$$

$$\alpha + \beta\sqrt{2}$$

$$\therefore \alpha + \beta = 168 + 56 = 224$$

27. If the sum of the first 20 terms of the series $\frac{4 \cdot 1}{4 + 3 \cdot 1^2 + 1^4} + \frac{4 \cdot 2}{4 + 3 \cdot 2^2 + 2^4} + \frac{4 \cdot 3}{4 + 3 \cdot 3^2 + 3^4} + \frac{4 \cdot 4}{4 + 3 \cdot 4^2 + 4^4} + \dots$ is $\frac{m}{n}$, where m and n are coprime, then m + n is equal to :

यदि श्रेणी $\frac{4 \cdot 1}{4 + 3 \cdot 1^2 + 1^4} + \frac{4 \cdot 2}{4 + 3 \cdot 2^2 + 2^4} + \frac{4 \cdot 3}{4 + 3 \cdot 3^2 + 3^4} + \frac{4 \cdot 4}{4 + 3 \cdot 4^2 + 4^4} + \dots$ के प्रथम 20 पदों का योग $\frac{m}{n}$ है, जहाँ m

और n असहभाज्य हैं, तो $m + n$ बराबर है :

(A) 423

(B) 422

(C) 420

(D) 421

Ans. D

Sol.

$$S = \sum_{r=1}^{20} \frac{4r}{4 + 3r^2 + r^4}$$

$$S = \sum_{r=1}^{20} \frac{4r}{(r^4 + 4r^2 + 4) - r^2}$$

$$S = \sum_{r=1}^{20} \frac{4r}{(r^2 + 2)^2 - r^2}$$

$$S = 2 \sum_{r=1}^{20} \frac{(r^2 + r + 2) - (r^2 - r + 2)}{(r^2 + r + 2)(r^2 - r + 2)}$$

$$S = 2 \sum_{r=1}^{20} \left(\frac{1}{r^2 - r + 2} - \frac{1}{r^2 + r + 2} \right)$$

$$S = 2 \left(\frac{1}{2} - \frac{1}{422} \right) = \frac{420}{422} = \frac{210}{211}$$

28. Let $B = \begin{bmatrix} 1 & 3 \\ 1 & 5 \end{bmatrix}$ and A be a 2×2 matrix such that $AB^{-1} = A^{-1}$. If $BCB^{-1} = A$ and $C^4 + \alpha C^2 + \beta I = O$, then $2\beta - \alpha$ is equal to :

माना $B = \begin{bmatrix} 1 & 3 \\ 1 & 5 \end{bmatrix}$ है तथा एक 2×2 आव्यूह A के लिए $AB^{-1} = A^{-1}$ है। यदि $BCB^{-1} = A$ तथा $C^4 + \alpha C^2 + \beta I = O$ हैं, तो

$2\beta - \alpha$ बराबर है :

(A) 16

(B) 2

(C) 10

(D) 8

Ans. C

Sol.

$$B = \begin{bmatrix} 1 & 3 \\ 1 & 5 \end{bmatrix}$$

$$AB^{-1} = A^{-1}$$

$$\Rightarrow A^2 = B$$

Also, $BCB^{-1} = A$

$$\Rightarrow C = B^{-1}AB$$

$$\Rightarrow C^4 = (B^{-1}AB)(B^{-1}AB)(B^{-1}AB)(B^{-1}AB)$$

$$= B^{-1}A^4B$$

$$= B^{-1}B^2B$$

$$\Rightarrow C^4 = B^2$$

$$\text{Also, } C^2 = (B^{-1}AB)(B^{-1}AB)$$

$$= B^{-1}A^2B$$

$$= B^{-1}BB$$

$$\Rightarrow C^2 = B$$

$$\Rightarrow C^4 + \alpha C^2 + \beta I = 0$$

$$\Rightarrow B^2 + \alpha B + \beta I = 0$$

$$B^2 = \begin{bmatrix} 1 & 3 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 1 & 5 \end{bmatrix} = \begin{bmatrix} 4 & 18 \\ 6 & 28 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 4 & 18 \\ 6 & 28 \end{bmatrix} + \alpha \begin{bmatrix} 1 & 3 \\ 1 & 5 \end{bmatrix} + \begin{bmatrix} \beta & 0 \\ 0 & \beta \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\Rightarrow 4 + \alpha + \beta = 0$$

$$\text{and } 18 + 3\alpha = 0$$

$$\Rightarrow \alpha = -6$$

$$\Rightarrow \beta = 2$$

$$\Rightarrow 2\beta - \alpha = 10$$

29. The sum of the series $\frac{1}{1-3 \cdot 1^2+1^4} + \frac{2}{1-3 \cdot 2^2+2^4} + \frac{3}{1-3 \cdot 3^2+3^4} + \dots$ upto 10-terms is :

श्रेणी $\frac{1}{1-3 \cdot 1^2+1^4} + \frac{2}{1-3 \cdot 2^2+2^4} + \frac{3}{1-3 \cdot 3^2+3^4} + \dots$ 10 पदों तक का योग है :

(A) $\frac{45}{109}$

(B) $\frac{55}{109}$

(C) $-\frac{45}{109}$

(D) $-\frac{55}{109}$

Ans. D

Sol. General term of the sequence

$$T_r = \frac{r}{1-3r^2+r^4}$$

$$T_r = \frac{r}{r^4-2r^2+1-r^2}$$

$$T_r = \frac{r}{(r^2 - 1)^2 - r^2}$$

$$T_r = \frac{r}{(r^2 - r - 1)(r^2 + r - 1)}$$

$$T_r = \frac{\frac{1}{2}[(r^2 + r - 1) - (r^2 - r - 1)]}{(r^2 - r - 1)(r^2 + r - 1)}$$

$$= \frac{1}{2} \left[\frac{1}{r^2 - r - 1} - \frac{1}{r^2 + r - 1} \right]$$

Sum of 10 terms,

$$\sum_{r=1}^{10} T_r = \frac{1}{2} \left[\frac{1}{-1} - \frac{1}{109} \right] = \frac{-55}{109}$$

30. Let α and β be the roots of the equation $px^2 + qx - r = 0$, where $p \neq 0$. If p , q and r be the consecutive terms of a non constant G.P. and $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{3}{4}$ then the value of $(\alpha - \beta)^2$ is :

माना समीकरण $px^2 + qx - r = 0$, $p \neq 0$ के मूल α तथा β हैं। यदि p , q तथा r एक परिवर्तनीय (non constant) G.P. के क्रमागत पद हैं तथा $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{3}{4}$ है, तो $(\alpha - \beta)^2$ का मान है :

- (A) 9 (B) 8 (C) $\frac{80}{9}$ (D) $\frac{20}{3}$

Ans. C

Sol. $px^2 + qx - r = 0$

roots are α , β

$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{3}{4}$$

$$4(\alpha + \beta) = 3\alpha\beta$$

$$4\left(\frac{-q}{p}\right) = 3\left(\frac{-r}{p}\right)$$

$$4q = 3r$$

$$(p, q, r) \equiv (A, AR, AR^2)$$

$$4AR = 3AR^2$$

$$\Rightarrow R = \frac{4}{3}$$

$\Rightarrow$ quadratic equation will be

$$Ax^2 + \frac{4A}{3}x - \frac{16}{9}A = 0$$

$$9x^2 + 12x - 16 = 0$$

$$(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$$

$$= \left(-\frac{12}{9}\right)^2 - 4\left(-\frac{16}{9}\right) = \frac{80}{9}$$

31. If $\alpha \neq a, \beta \neq b, \gamma \neq c$ and $\begin{vmatrix} \alpha & b & c \\ a & \beta & c \\ a & b & \gamma \end{vmatrix} = 0$, then $\frac{a}{\alpha-a} + \frac{b}{\beta-b} + \frac{\gamma}{\gamma-c}$ is equal to :

यदि $\alpha \neq a, \beta \neq b, \gamma \neq c$ तथा $\begin{vmatrix} \alpha & b & c \\ a & \beta & c \\ a & b & \gamma \end{vmatrix} = 0$ है, तो $\frac{a}{\alpha-a} + \frac{b}{\beta-b} + \frac{\gamma}{\gamma-c}$ बराबर है :

(A) 1

(B) 0

(C) 3

(D) 2

Ans. B

Sol. $R_1 \rightarrow R_1 - R_2, R_2 \rightarrow R_2 - R_3$

$$\begin{vmatrix} \alpha-a & b-\beta & 0 \\ 0 & \beta-b & c-\gamma \\ a & b & \gamma \end{vmatrix} = 0$$

$$(\alpha-a)(\gamma(\beta-b) - b(c-\gamma)) - (b-\beta)(-a(c-\gamma)) = 0$$

$$\gamma(\alpha-a)(\beta-b) - b(\alpha-a)(c-\gamma) + a(b-\beta)(c-\gamma)$$

$$\frac{\gamma}{\gamma-c} + \frac{b}{\beta-b} + \frac{a}{\alpha-a} = 0$$

32. The number of terms of an A.P. is even; the sum of all the odd terms is 24, the sum of all the even terms is 30 and the last term exceeds the first by $\frac{21}{2}$. Then the number of terms which are integers in the A.P. is :

एक A.P. में पदों की संख्या सम है, सभी विषम पदों का योग 24 है, सभी सम पदों का योग 30 है और अंतिम पद, प्रथम पद से $\frac{21}{2}$ अधिक है। तो इस A.P. में पूर्णांक पदों की संख्या है :

- (A) 10 (B) 4 (C) 6 (D) 8

Ans. B

Sol. Let the number of terms be $2n$

$$T_1 + T_3 + T_5 + \dots + T_{2n-1} = 24$$

$$T_2 + T_4 + T_6 + \dots + T_{2n} = 30$$

$$(T_2 - T_1) + (T_4 - T_3) + \dots + (T_{2n} - T_{2n-1}) = 6$$

$$nd = 6$$

$$(a + (2n + 1)d) - a = \frac{21}{2}$$

$$\Rightarrow 2nd - d = \frac{21}{2}$$

$$\Rightarrow 12 - \frac{21}{2} = d$$

$$\Rightarrow d = \frac{3}{2}$$

$$\therefore n = 4$$

$$\therefore \text{Total terms} = 8$$

$$\text{Integral terms} = 4$$

33. If the set of all $a \in \mathbf{R}$, for which the equation $2x^2 + (a - 5)x + 15 = 3a$ has no real roots, is the interval (α, β) and

$X = \{x \in \mathbf{Z} : \alpha < x < \beta\}$, then $\sum_{x \in X} x^2$ is equal to :

यदि सभी $a \in \mathbf{R}$, जिनके लिए समीकरण $2x^2 + (a - 5)x + 15 = 3a$ का कोई वास्तविक मूल नहीं है, का समुच्चय अन्तराल (α, β)

है, और $X = \{x \in \mathbf{Z} : \alpha < x < \beta\}$ है, और $\sum_{x \in X} x^2$ बराबर है :

- (A) 2109 (B) 2119 (C) 2139 (D) 2129

Ans. C

Sol. $(a - 5)^2 - 8(15 - 3a) < 0$

$$a^2 + 14a + 25 - 120 < 0$$

$$a^2 + 14a - 95 < 0$$

$$(a + 19)(a - 5) < 0$$

$$a \in (-19, 5)$$

$$\therefore -19 < x < 5$$

$$\therefore \sum_{x \in X} x^2 = (1^2 + 2^2 + \dots + 4^2) + (1^2 + 2^2 + \dots + 18^2)$$

$$= \frac{4 \times 5 \times 9}{6} + \frac{18 \times 19 \times 37}{6}$$

$$= 30 + 2109$$

$$= 2139$$

34. Let α and β be the roots of $x^2 + \sqrt{3}x - 16 = 0$, and γ and δ be the roots of $x^2 + 3x - 1 = 0$. If $P_n = \alpha^n + \beta^n$ and

$Q_n = \gamma^n + \delta^n$, then $\frac{P_{25} + \sqrt{3}P_{24}}{2P_{23}} + \frac{Q_{25} - Q_{23}}{Q_{24}}$ is equal to :

माना समीकरण $x^2 + \sqrt{3}x - 16 = 0$ के मूल α और β हैं तथा समीकरण $x^2 + 3x - 1 = 0$ के मूल γ और δ हैं। यदि $P_n = \alpha^n + \beta^n$

और $Q_n = \gamma^n + \delta^n$ हैं, तो $\frac{P_{25} + \sqrt{3}P_{24}}{2P_{23}} + \frac{Q_{25} - Q_{23}}{Q_{24}}$ बराबर है :

(A) 4

(B) 7

(C) 5

(D) 3

Ans. C

Sol. $x^2 + \sqrt{3}x - 16 = 0$ $\begin{matrix} \nearrow \alpha \\ \searrow \beta \end{matrix}$

$$P_n + \sqrt{3}P_{n-1} - 16P_{n-2} = 0$$

$$P_{25} + \sqrt{3}P_{24} - 16P_{23} = 0$$

$$\frac{P_{25} + \sqrt{3}P_{24}}{2P_{23}} = 8$$

$$x^2 + 3x - 1 = 0$$
 $\begin{matrix} \nearrow \gamma \\ \searrow \delta \end{matrix}$

$$Q_{25} + 3Q_{24} - Q_{23} = 0$$

$$Q_{25} - Q_{23} = -3Q_{24}$$

$$\frac{Q_{25} - Q_{23}}{Q_{24}} = -3$$

35. If $\begin{vmatrix} x+1 & x & x \\ x & x+\lambda & x \\ x & x & x+\lambda^2 \end{vmatrix} = \frac{9}{8}(103x+81)$, then $\lambda, \frac{\lambda}{3}$ are the roots of the equation :

यदि $\begin{vmatrix} x+1 & x & x \\ x & x+\lambda & x \\ x & x & x+\lambda^2 \end{vmatrix} = \frac{9}{8}(103x+81)$ है, तो $\lambda, \frac{\lambda}{3}$ किस समीकरण के मूल है :

(A) $4x^2 - 24x + 27 = 0$

(B) $4x^2 - 24x - 27 = 0$

(C) $4x^2 + 24x + 27 = 0$

(D) $4x^2 + 24x - 27 = 0$

Ans. A

Sol. $\begin{vmatrix} x+1 & x & x \\ x & x+\lambda & x \\ x & x & x+\lambda^2 \end{vmatrix} = \frac{9}{8}(103x+81)$

Put $x = 0$

$$\begin{vmatrix} 1 & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda^2 \end{vmatrix} = \frac{9}{8} \times 81$$

$$\lambda^3 = \frac{3^6}{2^3}$$

$$\lambda = \frac{9}{2}$$

$$\frac{\lambda}{3} = \frac{3}{2}$$

Equation : $x^2 - \left(\frac{9}{2} + \frac{3}{2}\right)x + \frac{9}{2} \times \frac{3}{2} = 0$

$$4x^2 - 24x + 27 = 0$$

36. Let $a \in \mathbb{R}$ and A be a matrix of order 3×3 such that $\det(A) = -4$ and $A + I = \begin{bmatrix} 1 & a & 1 \\ 2 & 1 & 0 \\ a & 1 & 2 \end{bmatrix}$, where I is the identity

matrix of order 3×3 . If $\det((a+1)\text{adj}((a-1)A)) = 2^m 3^n$, $m, n \in \{0, 1, 2, \dots, 20\}$, then $m+n$ is equal to :

माना $a \in \mathbb{R}$ है तथा कोटि 3×3 के एक आव्यूह A के लिए $\det(A) = -4$ और $A + I = \begin{bmatrix} 1 & a & 1 \\ 2 & 1 & 0 \\ a & 1 & 2 \end{bmatrix}$ है, जहाँ I , कोटि 3×3 का

तत्समक आव्यूह है। यदि $\det((a+1)\text{adj}((a-1)A)) = 2^m 3^n$, $m, n \in \{0, 1, 2, \dots, 20\}$ है, तो $m + n$ बराबर है :

(A) 14

(B) 17

(C) 16

(D) 15

Ans. C

Sol. $A = \begin{bmatrix} 1 & a & 1 \\ 2 & 1 & 0 \\ a & 1 & 2 \end{bmatrix} - I = \begin{bmatrix} 0 & a & 1 \\ 2 & 0 & 0 \\ a & 1 & 1 \end{bmatrix}$

$$|A| = -4 \Rightarrow 2 - 2a = -4 \Rightarrow a = 3$$

$$|(a+1)\text{adj}(a-1)A| = |4\text{adj}3A|$$

$$= 4^3 |\text{adj}3A|$$

$$= 4^3 \times |3A|^{3-1} = 64|3A|^2$$

$$= 64 \times (3^3)^2 |A|^2$$

$$= 2^6 \times 3^6 \times 16$$

$$2^m \times 3^n = 2^{10} \times 3^6$$

$$\therefore m = 10, n = 6$$

$$\Rightarrow m + n = 16$$

37. Suppose that the number of terms in an A.P. is $2k$, $k \in \mathbb{N}$. If the sum of all odd terms of the A.P. is 40, the sum of all even terms is 55 and the last term of the A.P. exceeds the first term by 27, then k is equal to :

माना कि एक A.P. में पदों की संख्या $2k$, $k \in \mathbb{N}$ है। यदि उस A.P. के सभी विषम पदों का योग 40, सभी सम पदों का योग 55 और A.P. का अन्तिम पद, पहले पद से 27 अधिक है, तो k बराबर है :

(A) 5

(B) 4

(C) 6

(D) 8

Ans. A

Sol. Let A.P. $a, a + d, a + 2d, \dots$

$$T_1 + T_3 + T_5 \dots \text{upto } n \text{ term} = 40$$

$$a + a + 2d + a + 4d \dots = 40$$

$$n(a + (n-1)d) = 40 \quad \text{_____ (1)}$$

$$T_2 + T_4 + T_6 \dots \text{upto } n \text{ term} = 55$$

$$a + d + a + 3d + a + 5d \dots = 55$$

$$\frac{n}{2}\{2(a+d)+(n-1)2d\} = 55$$

$$n(a+nd) = 55 \quad \text{_____} (2)$$

$$\text{Given } l - a = 27$$

$$a + (2n-1)d - a = 27$$

$$(2n-1)d = 27 \quad \text{_____} (3)$$

$$\text{Equ. (2) - (1)}$$

$$nd = 15 \quad \text{_____} (4)$$

$$\text{from (3) and (4)}$$

$$d = 3 \text{ and } n = 5$$

38. The sum $1 + 3 + 11 + 25 + 45 + 71 + \dots$ upto 20 terms, is equal to :

$$1 + 3 + 11 + 25 + 45 + 71 + \dots \text{ 20 पदों तक का योग है :}$$

$$(A) 6982 \quad (B) 7240 \quad (C) 8124 \quad (D) 7130$$

Ans. B

$$\text{Sol. } S_n = 1 + 3 + 11 + 25 + 45 + 71 + \dots + T_{n-1} + T_n$$

$$S_n = 1 + 3 + 11 + 25 + 45 + \dots + T_{n-1} + T_n$$

$$0 = 1 + 2 + 8 + 14 + 20 + \dots - T_n$$

$$T_n = 1 + \frac{n-1}{2} [4 + (n-2)6]$$

$$T_n = 1 + (n-1)(2+3n-6)$$

$$T_n = 1 + (n-1)(3n-4)$$

$$T_n = 3n^2 - 7n + 5$$

$$S_{20} = \sum_{h=1}^{20} (3n^2 - 7n + 5) = 7240$$

39. Let $A = \begin{bmatrix} 2 & 1 & 2 \\ 6 & 2 & 11 \\ 3 & 3 & 2 \end{bmatrix}$ and $P = \begin{bmatrix} 1 & 2 & 0 \\ 5 & 0 & 2 \\ 7 & 1 & 5 \end{bmatrix}$. The sum of the prime factors of $|P^{-1}AP - 2I|$ is equal to :

माना $A = \begin{bmatrix} 2 & 1 & 2 \\ 6 & 2 & 11 \\ 3 & 3 & 2 \end{bmatrix}$ तथा $P = \begin{bmatrix} 1 & 2 & 0 \\ 5 & 0 & 2 \\ 7 & 1 & 5 \end{bmatrix}$ हैं। तो $|P^{-1}AP - 2I|$ के अभाज्य गुणनखण्डों का योग है :

(A) 27

(B) 26

(C) 66

(D) 23

Ans. B

Sol. $|P^{-1}AP - 2I|$
 $= |P^{-1}AP - 2P^{-1}P|$
 $= |P^{-1}(A - 2I)P|$
 $= |P^{-1}| |A - 2I| |P|$
 $= |A - 2I|$
 $= \begin{vmatrix} 0 & 1 & 2 \\ 6 & 0 & 11 \\ 3 & 3 & 0 \end{vmatrix} = 69$

So, Prime factor of 69 is 3 and 23

So, sum = 26

40. Let $\alpha \in (0, \infty)$ and $A = \begin{bmatrix} 1 & 2 & \alpha \\ 1 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix}$. If $\det(\text{adj}(2A - A^T) \cdot \text{adj}(A - 2A^T)) = 2^8$, then $(\det(A))^2$ is equal to :

माना $\alpha \in (0, \infty)$ है तथा $A = \begin{bmatrix} 1 & 2 & \alpha \\ 1 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix}$ है। यदि $\det(\text{adj}(2A - A^T) \cdot \text{adj}(A - 2A^T)) = 2^8$ है, तो $(\det(A))^2$ बराबर है :

(A) 49

(B) 16

(C) 1

(D) 36

Ans. B

Sol. $|\text{adj}(A - 2A^T) \cdot \text{adj}(2A - A^T)| = 2^8$
 $Q = 2A^T - A \Rightarrow Q^T = 2A - A^T = -P$
 $|\text{adj}(P)\text{adj}(Q)| \Rightarrow |PQ| = -2^4$
 $\Rightarrow |P|(-|P|) = -2^4 \Rightarrow |P| = 4 \text{ and } |Q| = -4$

$$|A - 2A^T| = 4$$

$$A - 2A^T = \begin{bmatrix} 1 & 2 & \alpha \\ 1 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix} - 2 \begin{bmatrix} 1 & 1 & 0 \\ 2 & 0 & 1 \\ \alpha & 1 & 2 \end{bmatrix} = \begin{bmatrix} -1 & 0 & \alpha \\ -3 & 0 & -1 \\ -2\alpha & -1 & -2 \end{bmatrix}$$

$$\Rightarrow |A - 2A^T| = 1 + 3\alpha = 4$$

$$\Rightarrow \alpha = 1 \Rightarrow |A| = -4$$

$$\Rightarrow |A|^2 = 16$$